MODULE 3 - ENVIRONMENTAL POLLUTION & WASTE MANAGEMENT
Environmental Pollution (Sources, Impacts, Corrective and Preventive measures, Relevant Environmental Acts, Case-studies): Surface and Ground Water Pollution; Noise pollution; Soil Pollution and Air Pollution. Waste Management: Bio-medical Wastes; Solid waste; Hazardous wastes; E-wastes; Industrial and Municipal Sludge.

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01	Environmental Pollution	**L2**	**CO3**
02	Surface and Ground Water Pollution	**L2**	**CO3**
03	Noise pollution, Soil Pollution and Air Pollution	**L2**	**CO3**
04	Waste Management: Bio-medical Wastes	**L2**	**CO3**
05	Solid waste; Hazardous wastes	**L2**	**CO3**
06	E-wastes; Industrial and Municipal Sludge.	**L2**	**CO3**

Environmental Pollution

It is may be defined as any undesirable change in the physical, chemical or biological characteristics environment which can cause undesirable effects on various forms of life or property.

Types of pollutants

Biodegradable pollutants: That degrades or decomposes rapidly by natural processes.

Non-biodegradable pollutants: Do not decompose or decompose very slowly in the environment.

Air Pollution

Air pollution may be defined as an atmospheric condition in which certain substances are present in concentrations which can cause undesirable effects on man and his environment.

Sources of Air pollution

1) Natural sources

Natural sources of pollution are those that are caused due to natural phenomena. Ex: Volcanic eruptions, Forest fires, Biological decay, Pollen grains, Marshes, Radioactive materials.

2) Artificial sources

Artificial sources are those which are created by man. Ex: Thermal power plants, Vehicular emissions, Fossil fuel burning, agricultural activities etc.

Classification of Air Pollutants

Depending on the form of pollutants present in the environment**,** they are classified as:
Primary pollutants and Secondary pollutants

Primary pollutants are those that are directly emitted in the atmosphere in the harmful form Ex: CO, NO, CO2, SO2 etc.

Secondary pollutants are those that are formed by reacting with other components or some basic component of the atmosphere to form new pollutants. Ex: Oxides of Nitrogen (NO2 or NO3) react with moisture in the atmosphere to give Nitric acid

<u>Sources and common effects of common air pollutants</u>

- Carbon monoxide: It is a colourless, odourless gas that is poisonous animals. It is formed by incomplete combustion of carbon containing fuels.

Source of carbonmonoxide is cigarette smoking and incomplete combustion of fossil fuels (more than 77% comes from motor vehicle exhaust)

Health effects include reduced ability of red blood cells to carry oxygen to body cells and tissues. This leads to headache and anemia. At high levels it causes coma, irreversible brain damage and death.

Nitrogen Dioxide: It is a reddish-brown irritating gas that causes photochemical smog. In the atmosphere, it gets converted into nitric acid (HNO3). It is caused by burning fossil fuels in industries and power plants.

Health effects include lung irritation and damage. Environmental effects involve acid deposition leading to damage of trees, lakes, soil and ancient monuments. NO2 can damage fabrics.

- Sulphur Dioxide: It is a colourless and irritating gas that is formed by combustion of sulphur containing fossil fuels such as coal and oil. In the atmosphere it is converted into Sulphuric acid which is a major component of acid deposition.

Health effects involve breathing problems for healthy people.
Environmental effects involve reduced visibility and acid deposition on trees, lakes, soils and monuments leading to their deterioration and adverse effect on aquatic life.

- Suspended Particulate Matter (SPM): Includes a variety of particles and droplets (aerosols) that can be suspended in atmosphere for short to long periods.

Harmful effects of air pollution:

1) Effects on Human Health
   - Sulphur oxides causes irritation of eyes, nose, throat; Damage to lungs, Acute and Chronic Asthma, Bronchitis and Emphysema
   - Nitrogen oxides causes chronic obstructive pulmonary diseases, infant and cardiovascular diseases
   - Carbon monoxide induces headaches, dizziness, loss of vision, decreased muscular coordination and severe effects on the baby of a pregnant woman
   - Photochemical smog causes respiratory problems and irritation to eyes
   - VOCs causes tiredness, drowsiness, dizziness, nausea, confusion, bone marrow disease, Skin cancers, leukemia, cardiovascular disease
   - Dust particles induces stuffy noses, sinusitis, sore throats, dry cough, burning eyes, chest pain, aggravated asthma and chronic bronchitis
   - Lead damages the brain and central nervous system, kidneys and brain
   - Mercury brings nervous disorders, insomnia, memory loss, excitability, irritation, tremor and minamata disease

2) Effects on Plants
   Air pollutants affect plants by entering through stomata and destroy chlorophyll
   - Damages the leaf structure and causes necrosis (dead areas of leaf), loss of chlorophyll content causing yellowing of leafs (Chlorosis) or down ward curling of leaf (epinasty) and dropping of leaves (abscission).
   - PAN (Peroxyacetyl nitrate) causes silvering of lower surface of leaf, damage to young and more sensitive leaves and suppressed growth
   - Ozone causes flecks on leaf surface, premature ageing, necrosis and bleaching

3) Effects on Materials
   - Mixing of air pollutants and moisture in air causes acid rains which in turn damages the buildings, monuments and statues.
   - They also affects the aquatic life specifically fishes
   - Ozone in the atmosphere can cause cracking of rubber.

Control Measures

The atmosphere has several built-in self-cleaning processes such as dispersion, gravitational settling, flocculation, absorption, rain-washout, etc to cleanse the atmosphere. However, control of contaminants at their source level is a desirable and effective method through preventive or control technologies. Some measures that can be adopted in this direction are:

• Using unleaded petrol • Using fuels with low sulphur and ash content • Encouraging people to use public transport, walk or use a cycle as opposed to private vehicles • Ensure that houses, schools, restaurants and playgrounds are not located on busy streets	• Plant trees along busy streets as they remove particulates, carbon dioxide and absorb noise • Industries and waste disposal sites should be situated outside the city preferably on the downwind of the city. • Catalytic converters should be used to help control emissions of carbon monoxide and hydrocarbons

Noise Pollution

Noise is defined as, "the unwanted, unpleasant or disagreeable sound that causes discomfort to all living beings". Statistically, ever since the industrial revolution, noise in the environment has been doubling every ten years.

Sound intensity is measured in decibels (dB) that is the tenth part of the longest unit Bel. One dB is the faintest sound that a human ear can hear.

Types of Noise: Environmental noise has been doubling every ten years. Noise is classified as:

- Industrial Noise
- Transport Noise and
- Neighbourhood noise

Industrial Noise: It is sound with a high intensity caused by industry machines. Such noise pollution is caused by machines from various factories, industries and mills. Noise from mechanical saws and pneumatic drills is unbearable and a nuisance to the public.

The Indian Institute of Oto-Rino Laryngology, Chennai reported that increasing industrial pollution damages the hearing ability by at least 20%.

Workers in steel industry, who work close to heavy industrial blowers, are exposed to 112dB for eight hours suffer from occupational pollution.

Transport Noise: Transport noise mainly consists of traffic noise from road, rail and aircraft. The number of automobiles on roads like scooters, cars, motor cycles, buses, trucks and diesel engine vehicles has increased enormously in the recent past further aggravating the problem of transport noise.

Noise levels in most residential areas in metropolitan cities are hovering around the border line due to increased vehicular noise pollution.

Neighborhood noise: includes disturbance from household gadgets and community. Common sources being musical instruments, TV, VCR, Radios, Transistors, Telephones

Effects of Noise pollution

Noise pollution affects both human and animal health. It leads to:

- Contraction of blood vessels
- Making skin pale
- Excessive adrenalin in the blood stream which is responsible for high blood pressure.
- Blaring sounds are known to cause mental distress
- Heart attacks, neurological problems, birth defects and abortion
- Muscle contraction leading to nervous breakdown, tension, etc
- The adverse reactions are coupled with a change in hormone content of blood, which in- turn increases heart beat, constriction of blood vessels, digestive spams and dilation of the pupil of the eye.
- Adverse effects health, work efficiency and behaviour. Noise pollution may cause damage to the heart, brain, kidneys, liver and may produce emotional disturbance.

- The most immediate and acute effect of noise is impairment of hearing that diminishes some part of the auditory system. Prolonged exposure to noise of certain frequency pattern leads to chronic damage to the inner ear.
- Impulsive noise may cause psychological and pathological disorders
- Ultrasonic sound can affect the digestive, respiratory, cardiovascular system and semicircular canals of the internal ear.
- The brain is adversely affected by loud and sudden noise by jets and airplanes. People are subjected to psychiatric illness.
- Recent reports suggest that blood is thickened by excessive noise.
- The optical system of human beings is also affected by noise pollution. Severe noise pollution causes: Pupullary dilation, Impairment of night vision and Decrease in rate of colour perception

Control measures:

1) Source control: This includes source modification such as acoustic treatment to machine surface, design changes, limiting operational timings, etc

2) Transmission path intervention: This includes containing the source inside a sound insulating enclosure, constructing a noise barrier or provision of sound absorbing materials alongthe path.

3) Receptor control: This includes protection of the receiver by altering the work scheduleor provision of personal protection devices such as ear plugs for operating noisy machinery. Themeasure may include dissipation and deflection methods.

Oiling: Proper oiling will reduce noise from the machine.

Preventive measures:

1.Prescribing noise limits for vehicular traffic
2.Ban on honking (usage of horns) in certain areas
3.Creation of silence zones near schools and hospitals
4.Redesigning buildings to make them noise proof
5.Reduction of traffic density in residential areas
6.Giving preference to mass public transport system.

Soil Pollution

Soil pollution is defined as, "contamination of soil by human and natural activities which may cause harmful effect on living organisms". Composition of soil is listed below:

Types, effects and sources of soil pollution:

Soil pollution mainly occurs due to the following:

1.Industrial wastes
2.Urban wastes
3.Agricultural practices
4.Radioactive pollutants
5.Biological agents

1) Industrial wastes - Disposal of Industrial wastes is the major problem for soil pollution Sources: Industrial pollutants are mainly discharged from various origins such as pulp and paper mills, chemical fertilizers, oil refineries, sugar factories, tanneries, textiles, steel, distilleries, fertilizers, pesticides, coal and mineral mining industries, drugs, glass, cement, petroleum and engineering industries etc.

Effect: These pollutants affect and alter the chemical and biological properties of soil. As a result, hazardous chemicals can enter into human food chain from the soil or water, disturb the biochemical process and finally lead to serious effects on living organisms.

2) Urban wastes - Urban wastes comprise of both commercial and domestic wastes consisting of dried sludge and sewage. All the urban solid wastes are commonly referred to as refuse.

Constituents of urban refuse: This refuse consists of garbage and rubbish materials like plastics, glasses, metallic cans, fibres, paper, rubbers, street sweepings, fuel residues, leaves, containers, abandoned vehicles and other discarded manufactured products. Urban domestic wastes though disposed off separately from industrial wastes, can still be dangerous. This happens because theyare not easily degraded.

3) Agricultural practices – Modern agricultural practices pollute the soil to a large extent. With the advancing agro-technology, huge quantities of fertilizers, pesticides, herbicides and weedicides are added to increase the crop yield. Apart from these farm wastes, manure, slurry, debris, soil erosion containing mostly inorganic chemicals are reported to cause soil pollution

4) Radioactive pollutants/ - Radioactive substances resulting from explosions of nuclear testing laboratories and industries giving rise to nuclear dust radioactive wastes, penetrate the soil andaccumulate giving rise to land/soil pollution.

Ex:

- Radio nuclides of Radium, Thorium, Uranium, isotopes of Potassium (K-40) and Carbon(C-14) are commonly found in soil, rock, water and air.

- Explosion of hydrogen weapons and cosmic radiations include neutron, proton reactions by which Nitrogen (N-15) produces C-14. This C-14 participates in Carbon metabolismof plants which is then into animals and human beings.

- Radioactive waste contains several radio nuclides such as Strontium90, Iodine-129, Cesium-137 and isotopes of Iron which are most injurious. Strontium get deposited inbones and tissues instead of calcium.

- Nuclear reactors produce waste containing Ruthenium-106, Iodine-131, Barium-140, Cesium-144 and Lanthanum-140 along with primary nuclides Sr-90 with a half life 28years and Cs-137 with a half life 30 years. Rain water carries Sr-90 and Cs-137 to be deposited on the soil where they are held firmly with the soil particles by electrostatic forces. All the radio nuclides deposited on the soil emit gamma radiations.

5) Biological agents – Soil gets a large amount of human, animal and bird excreta which constitutea major source of land pollution by biological agents.

Ex: Heavy application of manures and digested sludge can cause serious damage to plants withina few years

Control measures of soil pollution:
1) Soil erosion can be controlled by a variety of forestry and farm practices. Ex: Planting trees onbarren slopes

2) Contour cultivation and strip cropping may be practiced instead of shifting cultivationTerracing and building diversion channels may be undertaken.

3) Reducing deforestation and substituting chemical manures by animal wastes also helps arrest soilerosion in the long term.

4) Proper dumping of unwanted materials: Excess wastes by man and animals pose a disposal problem. Open dumping is the most commonly practiced technique. Nowadays, controlled tipping is followed for solid waste disposal. The surface so obtained is used for housing or sportsfield.

5) Production of natural fertilizers: Bio-pesticides should be used in place of toxic chemical pesticides. Organic fertilizers should be used in place of synthesized chemical fertilizers. Ex:Organic wastes in animal dung may be used to prepare compost manure instead of throwing them wastefully and polluting the soil.

6) Proper hygienic condition: People should be trained regarding sanitary habits. Ex: Lavatories should be equipped with quick and effective disposal methods.

7) Public awareness: Informal and formal public awareness programs should be imparted toeducate people on health hazards by environmental education.
Ex: Mass media, Educational institutions and voluntary agencies can achieve this.

8) Recycling and Reuse of wastes: To minimize soil pollution, the wastes such as paper, plastics, metals, glasses, organics, petroleum products and industrial effluents etc should be recycled and reused. Ex: Industrial wastes should be properly treated at source. Integrated waste treatment methods should be adopted.

9) Ban on Toxic chemicals: Ban should be imposed on chemicals and pesticides like DDT, BHC,etc which are fatal to plants and animals. Nuclear explosions and improper disposal of radioactive wastes should be banned.

Water Pollution

Water pollution may be defined as "the alteration in physical, chemical and biologicalcharacteristics of water which may cause harmful effects on humans and aquatic life."Pollutants include:

1. Sewage
2. Industrial effluents and chemicals
3. Oil and other wastes

Chemicals in air dissolve in rain water, fertilizers, pesticides and herbicides leached from landpollute water.

Types, effects and sources of water pollution
Water pollution is any chemical, biological or physical change in water quality that has a harmful effect on livingorganisms or makes water unsuitable for desired uses.

1) Infectious agents
Ex: Bacteria, Viruses, Protozoa, and parasitic worms.

2) Oxygen demanding wastes (Dissolved oxygen): This degradation consumes dissolved oxygen inwater. Dissolved Oxygen (DO) is the amount of oxygen dissolved in a given quantity of water ata particular pressure and temperature.

The saturated point of DO varies from 8 to 15 mg/L Ex: Organic wastes such as animal manure and plant debris that can be decomposed by aerobic (oxygen-requiring) bacteria.

3) Human sources: Sewage, Animal feedlots, paper mills and food processing facilities.

Effects: Large populations of bacteria decomposing these wastes can degrade water quality by depleting water of dissolved oxygen. This causes fish and other forms of oxygen-consuming aquatic life to die.

4) Inorganic chemicals

Ex: Water soluble inorganic chemicals:

1. Acids
2. Compounds of toxic metals such as lead (Pb), arsenic (As) and selenium (Se)
3. Salts such as NaCl in oceans and fluoride (F^-) found in some soils

5) Organic chemicals

Ex: Oil, Gasoline, Plastics, Pesticides, Cleaning solvents and Detergents.

6) Plant nutrients

Ex: Water soluble compounds containing nitrate, Phosphate and Ammonium ions.

Control measures of water pollution

• Administration of water pollution control should be in the hands of state or centralgovernment • Scientific techniques should be adopted for environmental control of catchment areas of rivers, ponds or streams • Industrial plants should be based on recycling operations as it helps prevent disposal of wastes into natural waters but also extraction of products from waste. • Plants, trees and forests control pollution as they act as natural air conditioners. • Trees are capable of reducing sulphur dioxide and nitric oxide pollutants and hence more trees should be planted.	• No type of waste (treated, partially treated or untreated) should be discharged into any natural water body. Industries should develop closed loop water supply schemes and domestic sewage must be used for irrigation. • Qualified and experienced people must be consulted from time to time for effective control of water pollution. • Public awareness must be initiated regarding adverse effects of water pollution using the media. • Laws, standards and practices should be established to prevent water pollution and theselaws should be modified from time to time based on current requirements and technological advancements. • Basic and applied research in public health engineering should be encouraged.

Solid Waste Management

Rapid population growth and urbanization in developing countries has led to people generating enormous quantities of solid waste and consequent environmental degradation.

The waste is normally disposed in open dumps creating nuisance and environmental degradation. Solid wastes cause a major risk to public health and the environment. Management of solid wastes is important in order to minimize the adverse effects posed by their indiscriminate disposal.

Types of solid wastes: Depending on the nature of origin, solid wastes are classified into
1. Urban or municipal wastes
2. Industrial wastes and
3. Hazardous wastes

Sources of urban wastes
Urban wastes include the following wastes:

1) Domestic wastes containing a variety of materials thrown out from homes
Ex: Food waste, Cloth, Waste paper, Glass bottles, Polythene bags, Waste metals, etc.

2) Commercial wastes: It includes wastes coming out from shops, markets, hotels, offices, institutions, etc. Ex: Waste paper, packaging material, cans, bottle, polythene bags, etc.

3) Construction wastes: It includes wastes of construction materials. Ex: Wood, Concrete, Debris,etc.

4) Biomedical wastes: It includes mostly waste organic materials Ex: Anatomical wastes, Infectiouswastes, etc.

Classification of urban wastes

1) Bio-degradable wastes: Those wastes that can be degraded by microorganisms are called bio-degradable wastes
   Ex: Food, vegetables, tea leaves, dry leaves, etc.

2) Non-biodegradable wastes:_Urban solid waste materials that cannot be degraded by microorganisms are called non-biodegradable wastes.
   Ex: Polythene bags, scrap materials, glass bottles, etc.

Sources of industrial wastes

The main source of industrial wastes are chemical industries, metal and mineral processing industries.

Ex:

- Nuclear plants: It generated radioactive wastes
- Thermal power plants: It produces fly ash in large quantities
- Chemical Industries: It produces large quantities of hazardous and toxic materials.
- Other industries: Other industries produce packing materials, rubbish, organic wastes, acid,alkali, scrap metals, rubber, plastic, paper, glass, wood, oils, paints, dyes, etc.

Effect of improper solid waste management

1. Due to improper disposal of municipal solid waste on the roads and immediate surroundings,biodegradable materials undergo decomposition producing foul smell and become a breedingground for disease vectors.

2. Industrial solid wastes are the source for toxic metals and hazardous wastes that affect soilcharacteristics and productivity of soils when they are dumped on the soil

3. Toxic substances may percolate into the ground and contaminate the groundwater.

4. Burning of industrial or domestic wastes (cans, pesticides, plastics, radioactive materials andbatteries) produce furans, dioxins and polychlorinated biphenyls that are harmful to human beings.

Solid waste management involves waste generation, mode of collection, transportation,segregation of wastes and disposal techniques.

Steps involved in solid waste management:

Two important steps involved in solid waste management are:
Reduce, Reuse and Recycle of Raw Materials

Reduce - If usage of raw materials is reduced, the generation of waste also gets reduced
Reuse - Refillable containers that are discarded after use can be reused
Recycle- Recycling is the reprocessing of discarded materials into new useful products

Ex: Old aluminium cans and glass bottles are melted and recast into new cans and bottles, Preparation of cellulose insulation from paper, Preparation of automobile body and construction material from steel cans

This method 3R's help save money, energy, raw materials and reduces pollution.

Discarding Wastes:

The following methods are adopted for discarding wastes:

1. Land fill
2. Incineration and
3. Composting

1) Land fill: Solid wastes are placed in a sanitary landfill in which alternate layers of 80 cm thick refuse is covered with selected earth-fill of 20 cm thickness. After 2-3 years solid waste volume shrinks by 25-30% and land is used for parks, roads and small buildings. This is the most common and cheapest cheapest method of waste disposal and is mostly employed in Indian cities.

Advantages:

- It is simple and economical Segregation of wastes is not required
- Landfilled areas can be reclaimed and used for other purposes Converts low-lying, marshy waste-land into useful areas.
- Natural resources are returned to soil and recycled.

Disadvantages:

- Large area is required
- Land availability is away from the town, tansportation costs are high Leads to bad odour if landfill is not properly managed.
- Land filled areas will be sources of mosquitoes and flies requiring application of insecticides and pesticides at regular intervals.
- Causes fire hazard due to formation of methane in wet weather.

2) Incineration: It is a hygenic way of disposing solid waste. It is suitable if waste contains more hazardous material and organic content. It is a thermal process and very effective for detoxification of all combustible pathogens. It is expensive when compared to composting or land-filling. In this method municipal solid wastes are burnt in a furnace called incinerator.

Combustible substances such as rubbish, garbage, dead organisms and non-combustible matter such as glass, porcelain and metals are separated before feeding to incinerators. The non-combustible materials can be left out for recycling and reuse.

The leftover ashes and clinkers may account for about 10 to 20% which need further disposal by sanitary landfill or some other means.The heat produced in the incinerator during burning of refuse is used in the form of steam power for generation of electricity through turbines.

Municipal solid waste is generally wet and has a high calorific value. Therefore, it has to be dried first before burning. Waste is dried in a preheater from where it is taken to a large incinerating furnace called "destructor" which can incinerate about 100 to 150 tonnes per hour. Temperature normally maintained in a combustion chamber is about 700 C which may be increased to 1000 C when electricity is to be generated.

Advantages

- Residue is only 20-25% of the original and can be used as clinker after treatment requires very little space
- Cost of transportation is not high if the incinerator is located within city limits Safest from hygenic point of view
- An incinerator plant of 3000 tonnes per day capacity can generate 3MW of power.

Disadvantages

- Its capital and operating cost is high Operation needs skilled personnel
- Formation of smoke, dust and ashes needs further disposal and that may cause air pollution.

3) Composting : It is another popular method practiced in many cities in our country. In this method, bulk organic waste is converted into fertilizer by biological action.

Separated compostable waste is dumped in underground trenches in layers of 1.5m and finally covered with earth of 20cm and left for decomposition. Sometimes, action mycetes are introduced for active decomposition. Within 2 to 3 days biological action starts.

Organic matter is destroyed by action mycetes and lot of heat is liberated increasing the temperature of compost by 75C and the refuse is finally converted into powdery brown coloured odourless mass called humus that has a fertilizing value and can be used in agriculture.

Humus contains lot of Nitrogen essential for plant growth apart from phosphates and other minerals.

Advantages

- Manure added to soil increases water retention and ion-exchange capacity of soil. This method can be used to treat several industrial solid wastes.
- Manure can be sold thereby reducing cost of disposing wastes Recycling can be done

Disadvantages

- Non-consumables have to be disposed separately. The technology has not caught-up with the farmers and hence does not have an assured market.

E-Waste Management

E-waste (electrical and electronic waste) is the term used to describe old or discarded appliances that includes computers, consumer electronic goods etc.

It also refers to electronic products nearing the end of their useful life. Roughly 40 million metric tonnes of e-waste is produced globally each year, and only about 13% of that weight is recycled (that too mostly in developing countries).

Characteristics of E-Wastes

- Precious metals: Gold, Silver, Palladium and less quantities of platinum.
- Base metals: Copper, aluminum, nickel, tin, zinc, iron etc.
- Special metals: Indium, Bismuth, antimony, Selenium etc.
- Hazardous metals: Mercury, Beryllium, lead, cadmium, Arsenic etc.

Disposal methods

It is a fact that e-waste comprises of hazardous elements which offer the potential of increasing the intensity of their discharge in environment.

Based on composition following methods are suggested.

Land-filling: It is not possible to quantity environmental impacts from e-waste in landfills. They contain mixtures of various waste streams, and Emission of pollutants from landfills can be delayed for several years. As a consequence of the complex material mixture in e-waste, it is not possible to exclude environmental (long-term) risks even in secured land filling.

Incineration: Advantage of incineration of e-waste is the reduction of waste volume and the utilization of the energy of the combustible materials. By incineration some environmentally hazardous organic substances are converted into less hazardous compounds. A disadvantage of this also includes the emission of poisonous gases and residues into the atmosphere.

Impacts of E-Wastes and its ill-management

- The informal sector's recycling practices magnify health risks.
- Primary and secondary exposure to toxic metals (such as lead), results from open-air burning used to retrieve valuable components (such as gold).
- Combustion from burning e-waste creates fine particulate matter which is linked to pulmonary and cardiovascular disease.
- Guiyu in China is one of the largest e-waste recycling site in the world. It exhibits substantial digestive, neurological, respiratory and bone problems among workers.

Industrial and municipal sludge

They are byproducts of various processes in wastewater treatment plants, and they pose significant environmental management challenges. Each type of sludge has different characteristics and requires specific treatment and disposal methods.

1. Industrial Sludge

Definition:

Industrial sludge is a semi-solid residue generated from the treatment of industrial wastewater. It consists of various organic and inorganic materials, including toxic chemicals, heavy metals, and other hazardous substances, depending on the industrial process.

Sources:

Industrial sludge is produced in industries such as chemical manufacturing, pharmaceuticals, textiles, paper and pulp, metal processing, and food processing. Each industry generates sludge with different compositions, often containing contaminants specific to the industry.

Characteristics:

- Chemical Composition: Industrial sludge can contain heavy metals (like lead, mercury, and cadmium), organic pollutants (such as oils, greases, and phenols), and other hazardous chemicals.

- Physical Properties: It may vary in consistency from liquid to semi-solid, depending on the water content and the type of industrial process.

- Toxicity: The sludge can be highly toxic and hazardous, requiring careful handling, treatment, and disposal.

Treatment Methods:

- Chemical Treatment: Neutralization, precipitation, and coagulation are often used to remove hazardous substances.
- Biological Treatment: In some cases, biological processes like anaerobic digestion can be used to treat organic components of the sludge.
- Thermal Treatment: Incineration can be employed to reduce the volume of the sludge and destroy organic pollutants, though this requires careful management of emissions.
- Stabilization: Processes like lime stabilization or thermal drying can reduce the pathogenicity and odor of the sludge.

Disposal**:**

- Landfilling: Stabilized industrial sludge can be disposed of in hazardous waste landfills, but this requires compliance with strict environmental regulations.
- Land Application: In rare cases, if the sludge is treated and non-toxic, it can be used as a soil conditioner, though this is uncommon due to potential risks.
- Recycling: Some components of industrial sludge can be recovered and reused, for example, metals or energy from incineration.

2. Municipal Sludge

Definition:
Municipal sludge is the byproduct of sewage treatment in municipal wastewater treatment plants. It contains a mixture of organic and inorganic materials, including nutrients, pathogens, and other contaminants.

Sources:
Municipal sludge is generated from the treatment of domestic wastewater, including human waste, food waste, soaps, and detergents, as well as small amounts of industrial wastewater that enter the municipal sewage system.

Characteristics:

- Chemical Composition: Municipal sludge typically contains organic matter, nutrients (such as nitrogen and phosphorus), pathogens, and trace amounts of heavy metals and other contaminants.
- Physical Properties: Like industrial sludge, it can vary in consistency from liquid to semi-solid. Municipal sludge is generally less hazardous but can contain harmful pathogens.
- Nutrient Content: It often has high levels of nutrients, which can be beneficial for agriculture if properly treated.

Treatment Methods:

- Thickening: The water content of the sludge is reduced through processes like gravity thickening, flotation, or centrifugation.
- Stabilization: Biological processes such as anaerobic digestion or composting can reduce pathogens and odors while stabilizing the organic matter.
- Dewatering: Mechanical processes like belt presses, filter presses, or centrifuges are used to further reduce water content.
- Disinfection: Sludge is treated to destroy pathogens, often using lime stabilization, heat treatment, or chemical disinfection.
- Advanced Treatment: Techniques like thermal drying, pelletization, or incineration can be used to further process the sludge.

Disposal:

- Land Application: Treated municipal sludge, often called biosolids, can be used as a fertilizer or soil conditioner due to its nutrient content. However, this requires careful monitoring to avoid contamination of soil and water resources.
- Landfilling: Dewatered and stabilized sludge can be disposed of in landfills, although this is becoming less common due to environmental concerns and landfill space limitations.
- Incineration: Incinerating sludge reduces its volume significantly and can generate energy, but it requires proper emission controls to prevent air pollution.
- Composting: Composting municipal sludge with other organic materials can produce a soil amendment that is safe and beneficial for agricultural use.

Multiple Choice Questions (MCQ)

1) The word 'Environment' is derived from
a) Greek b)French c) Spanish d) English

2) Which among the following is a climatic factor?
a) pressure b) humidity c) temperature **d) all of the above**

3) The major atmospheric gas layer in stratosphere is
a) Hydrogen b) Carbon dioxide **c) Ozone** d) Helium

4) Which atmospheric sphere is closest to the earth surface?
a) **Troposphere** b) stratosphere c) mesosphere d) Exosphere

5) Which of the following is the terrestrial ecosystem?
a) Forest b) grass land c) desert **d) all of the above**

6) Ecological pyramids are studies of
a) Pyramid of numbers b) pyramid of biomass
c) Pyramid of energy **d) all of the above**

7) World Environment day is on
a) 5th May b) **5th June** c) 18th July d) 16th August

8) Factors responsible for balanced ecosystem are
a) balance between predator and prey
b) balance between vegetation, herbivorous and carnivorous
c) balance between competing species and biotic factors
d) All of the above

9) Which of the following is absorbed by green plants from the atmosphere?
a) Carbon dioxide b) Water c) Nutrients d) All of above

10) Habitat refers to
a) Physical conditions of the place where organisms live c) **Both a & b**
b) Chemical conditions of the place where organisms live d) None of a or b

11) Essential component of social security are
a) Meeting personal growth and development b) **Maintaining natural capital**
b) Fairness in distribution of costs of resources. d) Community resilience

12) Socio – economic security in environmental aspects involves
a) Fairness & equity in distribution costs for complete existing generation
b) Welfare of the present generation
c) Intra and inter-generational equity of resources
d) All of the above

13) A food web consists of
a) a portion of a food chain c) an organisms position in a food chain
b) Interlocking food chains d) a set of similar consumers

14) Which of the following statements are true?
a) Man is not dependent on nature
b) Resources are unlimited, so one can use them as per one's wish.
c) Energy can be converted from one form to another, but some % is lost
d) It need not be recycled or reused.

15) Which of the following conditions must be fulfilled to ensure food security?
a) Food must be available b) Each person must have access to it
b) Food utilized must fulfill nutritional requirements d) **all of the above**

16) 'Earth Day' is held every year on
a) June 5th b) November 23rd **c) April 22nd** d) January 26th

17) Eutrophication is
a) an improved quality of water in lakes
b) a process in carbon cycle
c) The result to accumulation of plant nutrients in water bodies
d) a water purification technique

18) Major purpose of most of the Dams around the world is
a) Power generation b) Drinking water supply c) Flood control **d) Irrigation**

19) Hot spot areas have:
a) Low density of biodiversity b) Only endangered plants
c) High density of hot springs d) **High density of biodiversity**

20) Plants use ……. gas for photosysthesis
a) Oxygen b) methane c) Nitrogen d) **Carbon dioxide**

21) Deforestation means
a) Forests Preservation b) agriculture
c) **Destruction of forests** d) monocrop cultivation

22) What percentage of its geographical area of a country should be under forest cover?
a) 23% b) 43% c) 13% d) **33%**

23) About ……% of the earth's surface is covered by water
a) 53% b) 19% c) **71%** d) 90%

24) During photosynthesis, trees produce
a) **Oxygen** b) Carbon Dioxide c) Nitrogen d) Carbon Monoxide

25) Forests prevent soil erosion by binding soil particles in their
a) stems b) **roots** c) leaves d)buds

26) Major causes of deforestation are
a) shifting cultivation b) Fuel requirements c) Raw materials for industries d) **All**

27) Which of the following statements about forest is not correct?
a) Forests reduces soil erosion b) Provides recreational opportunities
c) Provides a source of economic development d) **None of the above**

28) The average life expectancy around the world is currently
a) Decreasing b) **Increasing** c) Not changing d) Stabilizing

29) Which of the following is the problem not associated with population growth
a) Increase resource consumption b) Environmental pollution
c) Food and energy shortages d) **None of these**

30) The major objectives of family welfare programs in India is
a) Disease control b) **Population growth rate control**
c) Employment generation d) None of these

31) Population ageing is
a) The increase in the average age of the population
b) The result of decreased death and birth rates
b) The trend where more people live to reach old age while fewer children are born
c) **All of the above**

32) Which of the following is not population characteristic
a) Doubling time b) Total fertility rate
c) **Gross domestic product growth rate** d) Infant mortality rate

33) Demography is the study of
a) Animals behavior b) **Population growth**
c) River d) None of these

34) Which of the following is not a solution for global warming?
a) Reducing fossil fuel consumption b) Planting more trees
c) **De-forestation** d) None of the above

35) The meaning of global warming is
a) **Increase in the temp. of climate** b) A planet hotter than earth
c) Solar radiation d) Cooling effect

36) Global warming may bring about the following change in the climate of the earth
a) Increase in the rain fall b) Desertification
c) Drought d) **All of the above**

37) Which greenhouse gas is known as colourless, non-flammable, sweetish odour & laughing gas?
a) Methane b) CO_2 c) **Nitrous Oxide** d) Sulfur hexa fluoride

38) Forests are extremely important because they
a) Provide clean water and clean air b) Provide habitat for wild life
c) Provide recreation and a change from hectic urban life d) **all of the above**

39) Deforestation means
a) Creation of new forest land for the wild life b) planting trees in the cities
c) **Conversion of forest for agriculture / homes** d) not managing the forest properly

40) Sound beyond which of the following level can be regarded as a pollutant
a) 40dB b) **80dB** c) 120dB d) 150dB

41) Which of the following is a major source of thermal pollution of water bodies
a) Sewage treatment plants b) **Thermal power plants**
c) Solid waste disposal d) all

42) Smog is
a) A natural phenomenon c) Colorless
b) **combination of smoke and fog** d) all of the above

43) Air pollution from automobiles can be controlled by fitting
a) Electrostatic precipitator b) wet scrubber
c) **Catalytic converter** d) all of the above

44) Which of the following statements about carbon monoxide is true
a) Forms complex with haemoglobin b) Forms complex with leg-hemoglobin
c) form by incomplete combustion of fossil fuels d) **a and c**

45) Which of the following is a point source of water pollution
a) Factories b) Sewage treatment plants
c) Urban and Sub-Urban lands d) **a & b**